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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 6

Continuation

This lesson continues with more examples

Example . Given $y = \ln(3x + 1)^{17}$, find $\frac{dy}{dx}$

Solution: Let $u = 3x + 1$ then $\frac{du}{dx} = 3$

$$y = \ln u^{17} = 17 \ln u \text{ and } \frac{dy}{du} = \frac{17}{u}$$
$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \frac{17}{u} (3) = \frac{51}{u} = \frac{51}{3x+1}$$

□

Example . Find $\frac{dy}{dx}$ given $y = \ln \left\{ \frac{\sqrt{x^2+1}}{\sqrt[3]{x^3+1}} \right\}$

Solution: $y = \frac{1}{2} \ln(x^2 + 1) - \frac{1}{3} \ln(x^3 + 1)$

$$\frac{dy}{dx} = \frac{2x}{2(x^2+1)} - \frac{3x^2}{3(x^3+1)} = \frac{x}{(x^2+1)} - \frac{x^2}{(x^3+1)}$$
$$\frac{x^4+x-x^4-x^2}{(x^2+1)(x^3+1)} = \frac{x-x^2}{(x^2+1)(x^3+1)}$$





Example . Given $y = \sin(\ln x)$, find $\frac{dy}{dx}$

Solution: Let $u = \ln x$ then $\frac{du}{dx} = \frac{1}{x}$

and $y = \sin u$, then $\frac{dy}{du} = \cos u$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos u \left(\frac{1}{x} \right) = \frac{1}{x} \cos(\ln x)$$



EXERCISE 1.  Differentiate $2e^{3x} + e^{4x}$

EXERCISE 2.  Differentiate $2e^{\sin 3t}$

EXERCISE 3.  Differentiate e^{e^x}

Example . If $y = \sqrt{x-1}e^x \ln x$, find $\frac{dy}{dx}$

$$\begin{aligned} \text{Solution: } \frac{dy}{dx} &= \frac{1}{2}(x-1)^{-1/2}e^x \ln x + \sqrt{x-1}e^x \ln x + \frac{\sqrt{x-1}e^x}{x} \\ &= \frac{1}{2\sqrt{x-1}}e^x \ln x + \sqrt{x-1}e^x \ln x + \frac{\sqrt{x-1}e^x}{x} \end{aligned}$$



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$$\begin{aligned} &= e^x \left[\frac{1}{2\sqrt{x-1}} \ln x + \sqrt{x-1} \ln x + \frac{\sqrt{x-1}}{x} \right] \\ &= \frac{e^x}{2x\sqrt{x-1}} [x \ln x + 2x(x-1) \ln x + 2(x-1)] \\ &= \frac{e^x}{2x\sqrt{x-1}} [(x + 2x^2 - 2x) \ln x + 2x - 2] \\ &= \frac{e^x}{2x\sqrt{x-1}} [(2x^2 - x) \ln x + 2x - 2] \end{aligned}$$

Example . Find the gradient of the curve $y = \ln \sqrt{1 + \sin 2x}$ at point $x = \frac{\pi}{2}$

Solution: $y = \ln \sqrt{1 + \sin 2x}$ $\frac{dy}{dx} = \frac{1}{2} \left(\frac{2 \cos 2x}{1 + \sin 2x} \right) = \frac{\cos 2x}{1 + \sin 2x}$
at $x = \frac{\pi}{2}$, slope $\frac{dy}{dx} = \frac{\cos 2\pi}{1 + \sin 2\pi} = -1$

Example . Differentiate with respect to x , x^x

Solution: Let $y = x^x$, we introduce $\ln$, $\ln y = x \ln x$, differentiate implicitly



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$$\frac{1}{y} \frac{dy}{dx} = \ln x + \frac{x}{x} = \ln x + 1 \therefore \frac{dy}{dx} = y (\ln x + 1)$$

$$\frac{dy}{dx} = x^x (\ln x + 1)$$

□

Example  Differentiate with respect to x , $x^{\cos x}$

Solution: Let $y = x^{\cos x}$, we introduce $\ln$, $\ln y = \cos x \ln x$, differentiate implicitly

$$\frac{1}{y} \frac{dy}{dx} = -\sin x \ln x + \frac{\cos x}{x} \therefore \frac{dy}{dx} = y \left(\frac{\cos x}{x} - \sin x \ln x \right)$$

$$\frac{dy}{dx} = x^{\cos x} \left(\frac{\cos x}{x} - \sin x \ln x \right)$$

□

EXERCISE 4.  Differentiate the following with respect to x

(a) $a^x + b^x$

(b) $x^{\sin x}$

(c) $(\sin x)^x$

(d) $(x + x^2)^x$

(e) $\frac{x}{(x-1)(x+3)^2(x^2-1)}$

(f) $\sqrt{\frac{1}{(x^2-1)(3x-4)^4}}$





(g) $y^x = x$

(h) $x^y = \sin x$

(i) $(1+x)y = \sin^{1/2} x$

(j) $y^{\sin x} = \sqrt{x}$

6.1. Second and higher order derivatives

$$\frac{dy}{dx} = y' \quad \text{First derivative}$$

Differentiating the first derivative w.r.t x

$$\frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2y}{dx^2} = y'' \quad \text{Second derivative}$$

$$\Rightarrow \frac{d}{dx} \left(\frac{d^2y}{dx^2} \right) = \frac{d^3y}{dx^3} = y'''$$

$\vdots$

$$\frac{d}{dx} \left(\frac{d^{n-1}y}{dx^{n-1}} \right) = \frac{d^ny}{dx^n} = y^n \quad n^{\text{th}} \text{ derivative}$$

Example . If $y = 4x^3 - 6x^2 - 9x + 1$

Solution: $\frac{dy}{dx} = 12x^2 - 12x - 9$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} (12x^2 - 12x - 9) = 24x - 12$$



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$$\frac{d^3y}{dx^3} = \frac{d}{dx} (24x - 12) = 24$$

$$\frac{d^4y}{dx^4} = \frac{d}{dx} (24) = 0$$

□

If $\frac{dy}{dx}$ is found in terms of a parameter t , then $\frac{d^2y}{dx^2}$ requires differentiation with respect to x so,

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \div \frac{dx}{dt}$$

Example . If $x = t^3$, $y = t^2 - t$ find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$

Solution: $\frac{dx}{dt} = 3t^2$ and $\frac{dy}{dt} = 2t - 1$, $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = (2t - 1) \left(\frac{1}{3t^2} \right) =$

$$\frac{2t-1}{3t^2}$$

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{2t-1}{3t^2} \right) = \frac{2(3t^2) - 6t(2t-1)}{(3t^2)^2} = \frac{6t^2 - 12t^2 + 6t}{(3t^2)^2} = \frac{6t - 6t^2}{(3t^2)^2} =$$

$$\frac{2-2t}{3t^3}$$

$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \left(\frac{2-2t}{3t^3} \right) \left(\frac{1}{3t^2} \right) = \frac{2-2t}{9t^5}$$

□





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Example . The parametric equations of a curve are $x = e^t$, $y = \sin t$, find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ as functions of t . Hence show that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$.

Solution: $\frac{dx}{dt} = e^t$ and $\frac{dy}{dt} = \cos t$, $\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \cos t \left(\frac{1}{e^t} \right) = \frac{\cos t}{e^t}$

$$\frac{d}{dt} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{\cos t}{e^t} \right) = \frac{e^t(-\sin t) - \cos t(e^t)}{(e^t)^2} = \frac{-(e^t \sin t + e^t \cos t)}{(e^t)^2} = \frac{-(e^t \sin t + e^t \cos t)}{e^{2t}}$$
$$\frac{d^2y}{dx^2} = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \left(\frac{-(e^t \sin t + e^t \cos t)}{e^{2t}} \right) \left(\frac{1}{e^t} \right) = \frac{-(e^t \sin t + e^t \cos t)}{e^{3t}} = \frac{-(\sin t + \cos t)}{e^{2t}}$$
$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = e^{2t} \left(\frac{-(\sin t + \cos t)}{e^{2t}} \right) + e^t \left(\frac{\cos t}{e^t} \right) + \sin t = -\sin t - \cos t + \cos t + \sin t = 0$$

□

EXERCISE 5.  $y = 3(1 - \cos 2\theta)$, $x = 3(2\theta - \sin 2\theta)$ find



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the equations of the tangent and normal lines at $\theta = \frac{\pi}{4}$

More Revision Questions

a) Given that $x = \sec \theta + \tan \theta$ and $y = \csc \theta + \cot \theta$, show that

$$x + \frac{1}{x} = 2 \sec \theta \quad \text{and} \quad y + \frac{1}{y} = 2 \csc \theta$$

Find $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ in terms of θ and hence show that $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

b) Find $\frac{dy}{dx}$ at $t = 0$ if $x = \frac{1+t}{1-2t}$ and $y = \frac{1+2t}{1-t}$



Solutions to Exercises

Exercise 1. Let $y = 2e^{3x} + e^{4x}$, $\frac{dy}{dx} = 6e^{3x} + 4e^{4x}$

Exercise 1



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Exercise 2. Let $y = 2e^{\sin 3t}$,

$$\frac{dy}{dx} = (2) 3 \cos 3t e^{\sin 3t} =$$

Exercise 2



Exercise 3. Let $y = e^{e^x}$, Then $\ln y = e^x$

$$\frac{1}{y} \frac{dy}{dx} = e^x \quad \Rightarrow \quad \frac{dy}{dx} = ye^x \quad \text{or} \quad \frac{dy}{dx} = e^x e^{e^x}$$

Exercise 3



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Exercise 4. (a) $a^x \ln a + b^x \ln b$ (b) $\left(\frac{1}{x} \sin x + \cos x \ln x\right) x^{\sin x}$

(c) $(\sin x)^x (\ln \sin x + x \cot x)$

(d) $(x + x^2) \left[\ln(x + x^2) + \frac{1+2x}{1+x} \right]$

(e) $x \left[\frac{1}{x} - \frac{1}{x-1} - \frac{2}{x+3} + \frac{2x}{x^2-1} \right]$

(f) $\frac{6+4x-9x^2}{(x^2-1)^{3/2}(3x-4)^3}$ (g) $\ln y + x/y \frac{dy}{dx} = \frac{1}{x}$

(h) $\ln x \frac{dy}{dx} + y/x = \cot x, \frac{dy}{dx} = (\cot x - y/x) \frac{1}{\ln x}$

(i) $\frac{dy}{dx} = \left(\frac{1}{2} \cot x - 1/x+1\right) y$ (j) $\frac{dy}{dx} = \left(\frac{1}{2x} - \cos x \ln x\right) y/\sin x$

Exercise 4



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Exercise 5. $\frac{dx}{d\theta} = 6 - 6 \cos 2\theta$ and $\frac{dy}{d\theta} = 6 \sin 2\theta$, $\frac{dy}{dx} = \frac{dy}{d\theta} \frac{d\theta}{dx} =$

$$6 \sin 2\theta \left(\frac{1}{6 - 6 \cos 2\theta} \right) = \frac{6 \sin 2\theta}{6 - 6 \cos 2\theta}$$

$$\frac{dy}{dx} = \frac{\sin 2\theta}{1 - \cos 2\theta} \text{ at } \theta = \frac{\pi}{4}, \frac{dy}{dx} = \frac{\sin 2\left(\frac{\pi}{4}\right)}{1 - \cos 2\left(\frac{\pi}{4}\right)} = 1$$

gradient of the tangent at p is 1

$$\text{when } \theta = \frac{\pi}{4} \quad x = 3 \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right) = \frac{3\pi - 6}{2}$$

$$\theta = \frac{\pi}{4} \quad y = 3 \left(1 - \cos \frac{\pi}{2} \right) = 3$$

The equation of the tangent at $\left(\frac{3\pi - 6}{2}, 3 \right)$ is $\frac{y - 3}{x - \frac{3\pi - 6}{2}} = 1$, $y - 3 = x - \frac{3\pi - 6}{2}$, it crosses y axis at $\left(0, \frac{6 - 3\pi}{2} \right)$

The equation of the normal at $\left(\frac{3\pi - 6}{2}, 3 \right)$ is $\frac{y - 3}{x - \frac{3\pi - 6}{2}} = -1$, $y - 3 = \frac{3\pi - 6}{2} - x$.

Exercise 5